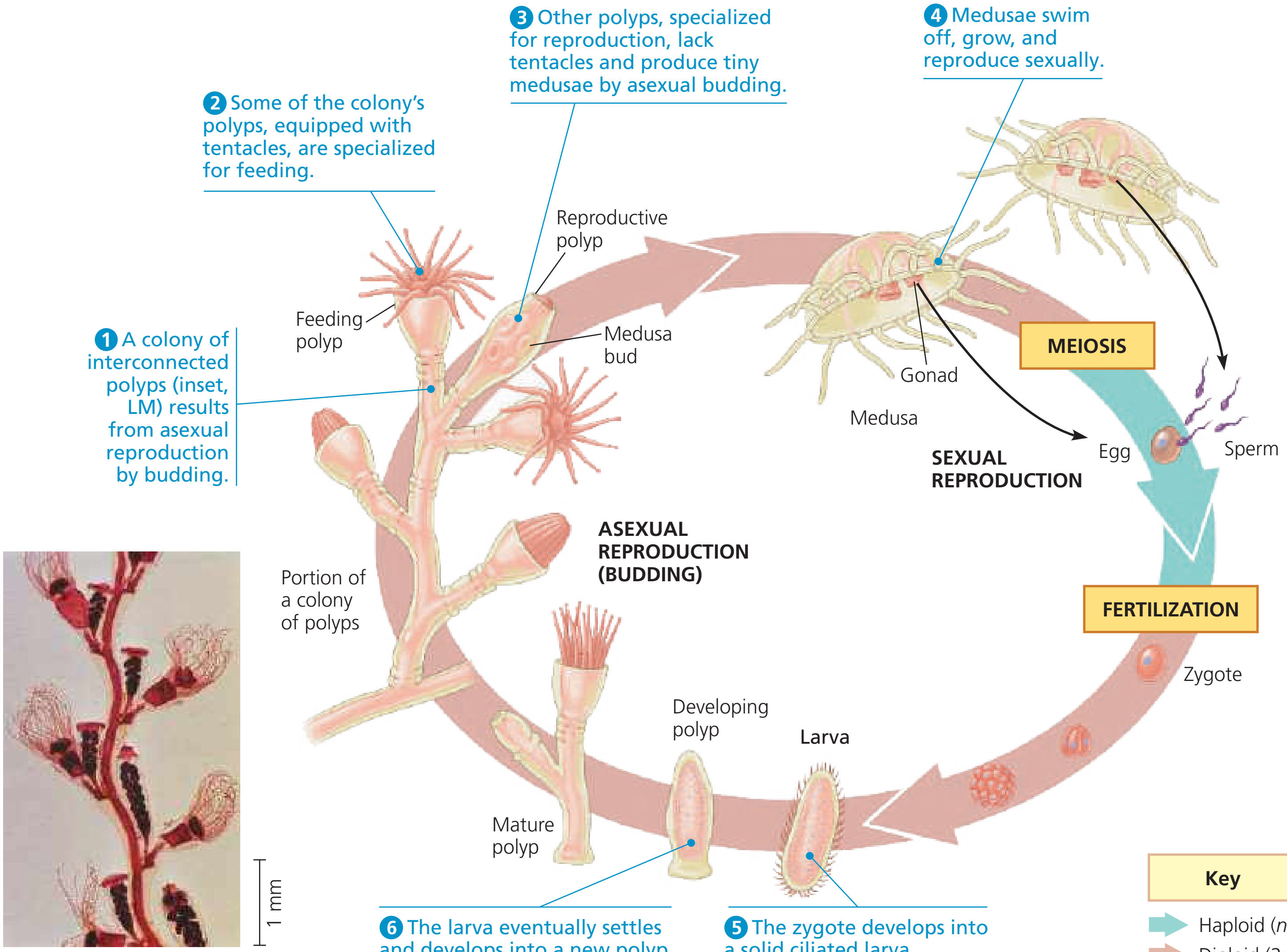
▼ **Figure 33.7 The life cycle of the hydrozoan *Obelia*.** The polyp is asexual, and the medusa is sexual, releasing eggs and sperm. These two stages alternate, one producing the other.



MAKE CONNECTIONS *Compare and contrast the* Obelia *life cycle to the life cycles in Figure 13.6. Which life cycle in that figure is most similar to that of* Obelia*? Explain. (See also Figure 29.5.)*

Anthozoans

Sea anemones and corals belong to the clade Anthozoa (see Figure 33.6b). These cnidarians occur only as polyps. Corals live as solitary or colonial forms, often forming symbioses with algae. Many species secrete a hard **exoskeleton** (external skeleton) of calcium carbonate. Each polyp generation builds on the skeletal remains of earlier generations, constructing rocklike reefs with shapes characteristic of their species. These skeletons are what we usually think of as coral.

Coral reefs are to tropical seas what rain forests are to tropical land areas: They provide habitat for many other species. Unfortunately, these reefs are being destroyed at an alarming rate. Pollution, overharvesting, and ocean acidification (see Figure 3.12) are major threats. Climate change is also contributing to their demise by raising seawater temperatures above the range in which corals thrive (see Concept 56.4).

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CONCEPT CHECK 33.2

1. Compare and contrast the polyp and medusa forms of cnidarians.
2. **VISUAL SKILLS** Use the cnidarian life cycle diagram in Figure 33.7 to determine the ploidy of a feeding polyp and of a medusa.
3. **MAKE CONNECTIONS** Many new animal body plans emerged during and after the Cambrian explosion. In contrast, cnidarians today retain the same diploblastic, radial body plan found in cnidarians 560 million years ago. Are cnidarians therefore less successful or less "highly evolved" than other animal groups? Explain. (See Concepts 25.3 and 25.6.)

For suggested answers, see Appendix A.